



National  
Qualifications  
2019

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# **2019 Engineering Science**

## **National 5**

### **Finalised Marking Instructions**

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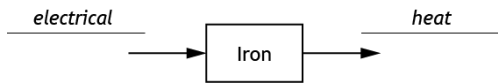
## General marking principles for National 5 Engineering Science

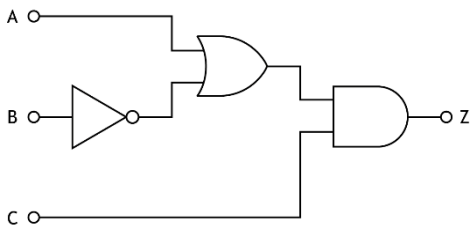
Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) Where a candidate makes an error at an early stage in a multi-stage calculation, credit should normally be given for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of nonmathematical reasoning.
- (c) All units of measurement will be presented in a consistent way, using negative indices where required (eg  $\text{ms}^{-1}$ ). Candidates may respond using this format, or solidus format (m/s) or words (metres per second), or any combination of these (eg metres/second).

## Marking instructions for each question

### Section 1

| Question |     |  | Expected response                                                                 | Max mark | Additional guidance                                                                                                                                                |
|----------|-----|--|-----------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) |  |  | 2        | 1 mark for electrical as input.<br>1 mark for heat as output.<br>Do not accept electricity/electric.                                                               |
|          | (b) |  | Closed loop (control)                                                             | 1        | Do not accept closed on its own.                                                                                                                                   |
|          | (c) |  |  | 1        | 1 mark for correct symbol.<br>-t° not required.<br>Small line must be parallel to long side of resistor.<br>Do not accept an arrow on any end.                     |
| 2.       |     |  | $P = \frac{E}{t}$ $P = \frac{6750}{270}$ $P = 25$ <b>P = 25 W (2 sf)</b>          | 2        | 1 mark for substitution.<br><br>1 mark for correct answer from given working with unit.                                                                            |
| 3.       | (a) |  | Air that operates a piston/cylinder.                                              | 1        | Descriptive response of function.<br>Accept instroke/outstroke/actuate a cylinder.<br>Must specifically refer to operating (stated or inferred) a cylinder/piston. |
|          | (b) |  | Air used to actuate a valve.                                                      | 1        | Descriptive response of function.<br>Must specifically refer to operating (stated or inferred) a valve.                                                            |

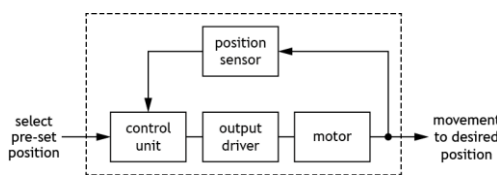
| Question |     |      | Expected response                                                                                                                                                                                                                                                       | Max mark | Additional guidance                                                                                                                                                                                                                         |
|----------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.       | (a) |      | $\text{Velocity Ratio} = \frac{\text{Speed of Input}}{\text{Speed of Output}}$ $\text{Velocity Ratio} = \frac{800}{320}$ $\text{Velocity Ratio} = 5 : 2$                                                                                                                | 2        | 1 mark for substitution.<br><br>1 mark for correct answer from given working.<br><br>Accept 2.5 : 1 or 2.5<br><br>Ignore any units.                                                                                                         |
|          | (b) |      | Anti-clockwise.                                                                                                                                                                                                                                                         | 1        | Do not accept other way or opposite way.<br><br>Accept indication of direction on diagram.                                                                                                                                                  |
| 5.       | (a) | (i)  | Civil.                                                                                                                                                                                                                                                                  | 1        |                                                                                                                                                                                                                                             |
|          |     | (ii) | Environmental.                                                                                                                                                                                                                                                          | 1        |                                                                                                                                                                                                                                             |
|          | (b) |      | Less stressful journeys .<br><br>Journey time quicker.<br><br>City roads less congested.<br><br>Areas bypassed will be quieter/safer.<br><br>City will be less polluted to live in.<br><br>Jobs created in the construction of the bypass/of housing in outlying areas. | 2        | 1 mark for each descriptive response of positive social impact of a city bypass.<br><br>Assume impact refers to bypass unless indicated in response.<br><br>Do not accept quick on its own.<br><br>Do not accept less congestion on bypass. |
| 6.       |     |      |                                                                                                                                                                                      | 3        | 1 mark for NOT gate wired to input B.<br><br>1 mark for OR gate wired to A and (NOT) B.<br><br>1 mark for AND gate wired to input C and A (OR) B.<br><br>No FTE from gate.                                                                  |

| Question |  |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                           |
|----------|--|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.       |  |  | <p>Positive:</p> <p>It does not produce CO<sub>2</sub> when in use.</p> <p>Reduces the need to burn fossil fuels.</p> <p>Does not contribute towards global warming.</p> <p>No pollution/clean when in use.</p> <p>Negative:</p> <p>Wildlife disrupted/habitats destroyed.</p> <p>Flow of water/flooding may erode the land/river bank.</p> <p>Unattractive pylons/dam site spoiling natural landscape.</p> <p>Vehicles noise when accessing the dam site.</p> | 2        | <p>1 mark for a description of an appropriate positive environmental impact when in use.</p> <p>1 mark for a description of an appropriate negative environmental impact when in use.</p> <p>Do not accept renewable/does not use fossil fuel on its own.</p> <p>Do not accept the same environmental impact of use as both positive and negative impact.</p> |

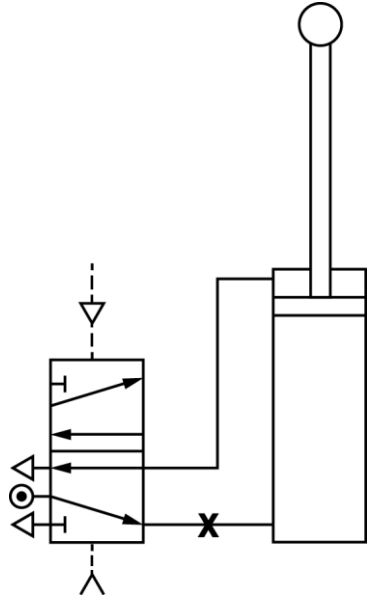
## Section 2

| Question |     |       | Expected response                                                                                                                                                                                                                                                      | Max mark | Additional guidance                                                                                                                                           |
|----------|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.       | (a) |       | $1200 + 270 = 1470 \text{ } (\Omega)$<br><br>$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$<br><br>$R_T = \frac{1470 \times 390}{1470 + 390}$<br><br>$R_T = 308.2$<br><br><b><math>R_T = 310 \text{ } \Omega \text{ (2 sf)}</math></b>                                       | 3        | 1 mark for series resistance (unit not required).<br><br><br>1 mark for substitution<br>Allow for FTE.<br><br>1 mark for answer with unit from given working. |
|          | (b) | (i)   | $V = IR$<br><br>$V = 0.031 \times 390$<br><br>$V = 12.09$<br><br><b><math>V = 12 \text{ V (2 sf)}</math></b>                                                                                                                                                           | 2        | 1 mark for substitution.<br><br>1 mark for answer with unit from given working with unit.                                                                     |
|          |     | (ii)  | $V = IR$<br><br>$12 = I \times 1470$<br><br>$I = \frac{12}{1470}$<br><br>$I = 0.00816$<br><br><b><math>I = 8.2 \text{ mA (2 sf)}</math></b>                                                                                                                            | 3        | 1 mark for substitution. Allow for FTE from (b)(i).<br><br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.           |
|          |     | (iii) | $A_3 = 0.0082 + 0.031$<br><br>$A_3 = 0.0392$<br><br><b><math>A_3 = 39 \text{ mA (2 sf)}</math></b><br><br><b>OR</b><br><br>$V = IR$<br><br>$12 = I \times 310$<br><br>$I = \frac{12}{310}$<br><br>$A_3 = 0.0387$<br><br><b><math>A_3 = 39 \text{ mA (2 sf)}</math></b> | 1        | 1 mark for answer with unit.<br><br>Allow for FTE from (b)(ii).<br><br><br><br><br><br><br><br><br><br>Allow for FTE from (a) and (b)(i).                     |

| Question |     |  | Expected response                                                                                                                                                                                                                                                                 | Max mark | Additional guidance                                                                                                                                                                                                                       |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          | (c) |  | <p>Graphene.<br/>Conducts electricity quickly.</p> <p>This would result in batteries charging quicker/circuits working quicker.</p> <p>Self driving car.<br/>Impact of emerging system is it may not be fully tested and so may have faults and this could cause an accident.</p> | 2        | <p>Accept impact for any emerging technology.</p> <p>1 mark for cause.</p> <p>1 mark for effect.</p> <p>If no technology named or the given example is clearly an established/developing; 1 mark maximum for cause <b>and</b> effect.</p> |

| Question |     |   | Expected response                                                                                                                                                                                                                                                                                                                                                              | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
|----------|-----|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------------------------------------------------------------------------------------------------------------------------------------|
| 9.       | (a) |   |                                                                                                                                                                                                                                                                                               | 1        | Dashed box around process.                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
|          | (b) |   | <p><i>The pre-set position is selected...</i></p> <p>The control unit compares the pre-set position with the actual position.</p> <p>The bed/motor will start to move (if it is not in the desired position).</p> <p>When the desired position is detected the motor will stop.</p>                                                                                            | 3        | <p>1 mark for comparison by control unit of positional sensor signal to pre-set value.</p> <p>1 mark for motor starting.</p> <p>1 mark for motor stopping in relation to desired position.</p>                                                                                                                                            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
|          | (c) |   | When an appropriate input signal is applied to the base an output current will flow at the collector/emitter.                                                                                                                                                                                                                                                                  | 2        | <p>1 mark for input signal at base (cause). Accept voltage as signal.</p> <p>1 mark for output current at collector/emitter based on given cause (effect).</p>                                                                                                                                                                            |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
|          | (d) |   | $Z = (\overline{A \cdot B}) + (A \cdot B)$                                                                                                                                                                                                                                                                                                                                     | 3        | <p>1 mark <math>(\overline{A \cdot B})</math> with brackets.</p> <p>1 mark <math>(A \cdot B)</math> with brackets.</p> <p>1 mark for OR-ing all statements.</p> <p>If just a single statement is given then bracket not required.</p> <p>accept alternatives such as</p> $Z = \overline{A \oplus B}$ $Z = \overline{(A+B)} + (A \cdot B)$ |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
|          | (e) |   | <table><tr><th>G</th><th>H</th><th>Y</th></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td></tr></table> | G        | H                                                                                                                                                                                                                                                                                                                                         | Y | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 3 | <p>1 mark per correct complete column.</p> <p>Allow for follow through error.</p> <p>Column H = NOT G..</p> <p>Column Y = F OR.</p> |
| G        | H   | Y |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 0        | 1   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 1        | 0   | 0 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |
| 1        | 0   | 1 |                                                                                                                                                                                                                                                                                                                                                                                |          |                                                                                                                                                                                                                                                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                     |

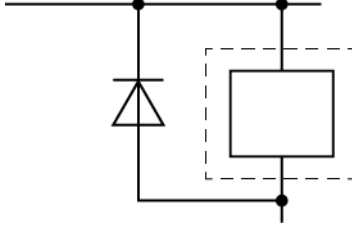


| Question |     |  | Expected response                                                                                                                                                                                                                                                                                 | Max mark | Additional guidance                                                                                                                                                                                                |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (a) |  | Saw blade: <b>rotary</b> .<br><br>Pneumatic piston: <b>reciprocating</b> .                                                                                                                                                                                                                        | 2        | 1 mark for saw blade motion.<br><br>1 mark for piston motion.<br><br>Accept rotational/reciprocal.<br><br>Do not accept rotating.                                                                                  |
|          | (b) |  | <i>When an increase in pressure is detected by valve 3...</i><br><br>...and when valve 1 is actuated (air is sent actuating valve 4), causing the piston to instroke...<br><br>valve 7 is actuated then there is a time delay...<br><br>...then valve 4 actuates causing the piston to outstroke. | 3        | 1 mark for AND control with valve 1 and 3 causing the piston to instroke.<br><br>1 mark for time delay after valve 7 being actuated.<br><br>1 mark piston outstroke after valve 4 actuated and valve 7/time delay. |
|          | (c) |  |                                                                                                                                                                                                                 | 1        | X on lower pipe to double acting cylinder.<br><br>Accept X on the bottom exhaust port of the 5/2 valve.                                                                                                            |
|          | (d) |  | The control of the piston movement could be easily changed/updated.<br><br>The piston time delay could be easily changed/shortened.<br><br>The piston time delay could be more consistent/accurate.                                                                                               | 1        | Descriptive response relating to advantage of microcontroller control of piston movement.<br><br>Do not accept fewer components or cost savings.                                                                   |

| Question |     |      | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                              | Max mark | Additional guidance                                                                                                                                                                                                                                                                                          |
|----------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (e) | (i)  | $A_{\text{piston}} = \frac{\pi d^2}{4}$ $A_{\text{piston}} = \frac{3 \cdot 14 \times 88^2}{4} = 6079 \text{ (mm}^2\text{)}$<br>$A_{\text{rod}} = \frac{\pi d^2}{4}$ $A_{\text{rod}} = \frac{3 \cdot 14 \times 24^2}{4} = 452 \cdot 2 \text{ (mm}^2\text{)}$<br>$A_{\text{effective}} = A_{\text{piston}} - A_{\text{rod}}$ $A_{\text{effective}} = 6079 - 452 \cdot 2$ $A_{\text{effective}} = 5627$ $A_{\text{effective}} = 5600 \text{ mm}^2 \text{ (2 sf)}$ | 3        | <p>1 mark for area of rod (unit not required).</p><br><p>1 mark for area of piston (unit not required).</p><br><p>1 mark for correct subtraction answer with unit.</p><br><p>Using the <math>\pi</math> function will give different intermediary values (6082 and 452.4) but same final answer to 2 sf.</p> |
|          |     | (ii) | $P = \frac{F}{A}$<br>$0 \cdot 20 = \frac{F}{5600}$<br>$F = 0 \cdot 20 \times 5600$<br>$F = 1120$<br>$F = 1 \cdot 1 \text{ kN (2 sf)}$                                                                                                                                                                                                                                                                                                                          | 3        | <p>1 mark for substitution<br/>Allow FTE from part (e)(i).</p><br><p>1 mark for transposition.</p><br><p>1 mark for correct answer from given working with unit.</p>                                                                                                                                         |

| Question |     |      | Expected response                                                                                                                                                                                                                                        | Max mark | Additional guidance                                                                                                                                                                                                                                                                                |
|----------|-----|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.      | (a) | (i)  | $\Sigma CWM = \Sigma ACWM$<br>$(R_A \times 2.4) = (680 \times 0.4) + (930 \times 3.6)$<br>$R_A = \frac{3620}{2.4}$<br>$R_A = 1508$<br><b><math>R_A = 1.5 \text{ kN (2 sf)}</math></b>                                                                    | 3        | 1 mark for substitution.<br><br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.                                                                                                                                                                           |
|          |     | (ii) | $\Sigma F_{\text{vertical}} = 0$<br>$930 + 680 = 1500 + R_B$<br>$R_B = 1610 - 1500$<br>$R_B = 110$<br><b><math>R_B = 110 \text{ N (2 sf)}</math></b>                                                                                                     | 2        | 1 mark for substitution<br>Allow FTE from part a(i).<br><br>1 mark for correct answer from given working with unit.<br><br>If moments used to determine $R_B$ then accepted 101.7 N .                                                                                                              |
|          | (b) |      | Equilibrium.                                                                                                                                                                                                                                             | 1        | Ignore additional description                                                                                                                                                                                                                                                                      |
|          | (c) |      | $\sigma = \frac{F}{A}$<br>$0.06 = \frac{2500}{A}$<br>$A = \frac{2500}{0.06}$<br>$A = 41670$<br><b><math>A = 42000 \text{ mm}^2 \text{ (2 sf)}</math></b>                                                                                                 | 3        | 1 mark for substitution..<br><br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.                                                                                                                                                                          |
|          | (d) |      | Design how to connect the visitor centre to the national grid.<br><br>Calculate the electrical power requirements of the visitor centre.<br><br>Select appropriately rated cables.<br><br>Calculate the lighting requirements/ plan the lighting layout. | 2        | 1 mark for any appropriate descriptive response of an engineer's <b>task</b> (design/calculate/select/model...) <b>and</b> the electrical <b>aspect</b> (must be linked to design of visitor centre).<br><br>Not design a circuit/wiring on its own.<br><br>Not electronic or electrician related. |

| Question |     |      | Expected response                                                                                                                                                                                         | Max mark | Additional guidance                                                                                                                                                                                                                                                            |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12.      | (a) | (i)  | $\text{time} = 60 + 20 = 80 \text{ (s)}$<br>$E_e = V I t$<br>$E_e = 230 \times 12 \times 80$<br>$E_e = 220800$<br>$E_e = 220 \text{ kJ (2 sf)}$                                                           | 3        | <p>1 mark for determining time in seconds. Unit not required.</p> <p>1 mark for substitution.<br/>Allow FTE for time.</p> <p>1 mark for correct answer from given working with unit.</p> <p>Accept calculating energy from power x time.</p>                                   |
|          |     | (ii) | $\eta = \frac{E_{\text{out}}}{E_{\text{in}}}$<br>$0.64 = \frac{E_{\text{out}}}{220000}$<br>$E_{\text{out}} = 0.64 \times 220000$<br>$E_{\text{out}} = 140800$<br>$E_{\text{out}} = 140 \text{ kJ (2 sf)}$ | 3        | <p>1 mark for substitution<br/>Allow for FTE from part a(i).</p> <p>1 mark for transposition..</p> <p>1 mark for correct answer from given working with unit</p>                                                                                                               |
|          | (b) |      | <p>Lubricate winch axles to reduce friction.</p> <p>Use a V belt so that there is less slippage/energy lost.</p>                                                                                          | 2        | <p>Explanation must relate to motorised winch modifications.</p> <p>1 mark for cause that increases the efficiency of the winch.</p> <p>Accept use of lubrication/bearings/toothed belt/V belt chain.</p> <p>1 mark for effect on friction/energy losses from given cause.</p> |
|          | (c) |      | 600 (N)                                                                                                                                                                                                   | 1        | Unit not required                                                                                                                                                                                                                                                              |
|          | (d) |      | $\varepsilon = \frac{\Delta l}{l}$<br>$0.0030 = \frac{\Delta l}{2700}$<br>$\Delta l = 0.003 \times 2700$<br>$\Delta l = 8.1$<br>$\Delta l = 8.1 \text{ mm (2 sf)}$                                        | 3        | <p>1 mark for substitution.</p> <p>1 mark for transposition.</p> <p>1 mark for correct answer from given working with unit.</p>                                                                                                                                                |

| Question |     |  | Expected response                                                                                                                                      | Max mark | Additional guidance                                                                                                                                                 |
|----------|-----|--|--------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13.      | (a) |  | <p>All lamps will have maximum brightness/230 volts.</p> <p>If one lamp fails the rest will remain on.</p>                                             | 2        | <p>Descriptive response.</p> <p>1 mark for each advantage.</p> <p>Do not accept all lamps will be the same brightness/voltage on its own.</p>                       |
|          | (b) |  |                                                                       | 2        | <p>1 mark for correct symbol (accept if line in centre of triangle is omitted).</p> <p>1 mark for correct orientation and connection. Allow FTE.</p>                |
|          | (c) |  | <p><i>As the light level decreases...</i></p> <p>...the resistance (of the LDR) increases..</p> <p>...voltage (<math>V_{in}</math>) will increase.</p> | 2        | <p>Descriptive response.</p> <p>1 mark for resistance of LDR increase.</p> <p>1 mark for input voltage increase.</p> <p>Apply FTE based on resistance response.</p> |
|          | (d) |  | $V_{in}$ will increase.                                                                                                                                | 1        | No FTE from 13(c).                                                                                                                                                  |

| Question |     |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                           | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13.      | (e) |  | $5 - 2.7 = 2.3 \text{ (V)}$<br><br>$\frac{R_1}{1.6} = \frac{2.3}{2.7}$<br><br>$R_1 = \frac{2.3}{2.7} \times 1.6$<br><br>$R = 1.363$<br><br><b>R = 1.4 kΩ (2 sf)</b><br><br><b>OR</b><br><br>$V_R = IR$<br>$2.7 = I \times 1.6$<br><br>$I = 1.688 \text{ (mA)}$<br><br>$V_R = 5.0 - 2.7 = 2.3 \text{ (V)}$<br><br>$V = IR$<br><br>$2.3 = 1.688 \times R_R$<br><br>$R = \frac{2.3}{1.688}$<br><br>$R = 1.363$<br><br><b>R = 1.4 kΩ (2 sf)</b> | 4        | <p>1 mark for voltage across variable resistor. Ignore unit.</p> <p>1 mark for substitution (apply FTE and accept 5.0 V if <math>V_1</math> is not calculated).</p> <p>1 mark for transposition.</p> <p>1 mark for correct answer from given working with unit.</p> <p>1 mark for calculating current (ignore unit).</p> <p>1 mark for calculating voltage <math>V_R</math> (ignore unit).</p> <p>1 mark for transposition (allow FTE).</p> <p>1 mark for correct answer from given working with unit.</p> |
| 13.      | (f) |  | 700 (lux)                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1        | Unit not required.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

| Question |     |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Max mark | Additional guidance                                                                                                                                                                                                                                                         |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13.      | (g) |  | <p>Quicker to develop solution.</p> <p>Easier to change circuit during testing.</p> <p>Reduces cost of designing as components will not be destroyed.</p> <p>Safer as 230 V circuit can be tested without fear of electrocution.</p> <p>No risk of damage to any actual components.</p>                                                                                                                                                                                                                                                           | 2        | <p>Descriptive advantage.</p> <p>1 mark for each relevant statement.</p> <p>Not faster, cheaper, easier or safer on its own.</p> <p>Do not accept testing can be carried out without an advantage.</p> <p>Cost responses must relate to speed or no damaged components.</p> |
| 14.      | (a) |  | <p>The harvester would be expensive to buy/maintain/run.</p> <p>Workers would lose their jobs therefore lose income.</p> <p>Harvester would be able to pick the fruit quicker/more fruit picked in a day meaning more profit.</p> <p>Increase profit for the farmer as wages would be reduced.</p> <p>Only ripe fruit would be picked therefore reducing waste increasing overall tonnage and profit.</p> <p>No need to house so many workers therefore increased profits.</p> <p>Workers would move away causing local shops to lose income.</p> | 2        | <p>1 mark for each appropriate economic description.</p> <p>Job and speed based responses must include reference to income/profit.</p>                                                                                                                                      |

| Question |     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14.      | (b) | $2200 \times 12 = \text{Output speed} \times 48$<br><br>$\text{Output speed} = \frac{26400}{48}$<br><br>$\text{Output speed} = 550 \text{ (revs min}^{-1}\text{)}$<br><br>$550 \times 16 = \text{Output speed} \times 24$<br><br>$\text{Output speed} = \frac{8800}{24}$<br><br>$\text{Output speed} = 366.7$<br><br><b>Output speed = 370 revs min<sup>-1</sup> (2 sf)</b><br><br>OR<br><br>$\frac{\text{output speed}}{\text{input speed}} = \frac{A}{B} \times \frac{C}{D}$<br><br>$\frac{\text{output speed}}{2200} = \frac{12}{48} \times \frac{16}{24}$<br><br>$\text{output speed} = \frac{12}{48} \times \frac{16}{24} \times 2200$<br><br>$\text{Output speed} = \frac{1}{6} \times 2200$<br><br>$\text{Output speed} = 366.7$<br><br><b>Output speed = 370 revs min<sup>-1</sup> (2 sf)</b> | 4        | 1 mark for substitution.<br><br>1 mark for correct answer from given working (unit not required).<br><br>1 mark for substitution<br>Allow FTE.<br><br>1 mark for correct answer from given working with unit.<br><br>Do not accept RPM.<br><br>1 mark for first ratio.<br><br>1 mark for second ratio.<br><br>1 mark for transposition (2200/6 if ratios inverted).<br><br>1 mark for answer from working with unit. |



| Question |     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15.      | (a) | <pre> graph TD     Start([Start]) --&gt; Pin0On{pin 0 on?}     Pin0On -- No --&gt; Start     Pin0On -- Yes --&gt; Pin7On[/pin 7 on/]     Pin7On --&gt; Pin6On[/pin 6 on/]     Pin6On --&gt; Wait08[Wait 0.8 s]     Wait08 --&gt; Pin6Off[/pin 6 off/]     Pin6Off --&gt; Pin5On[/pin 5 on/]     Pin5On --&gt; Wait03_1[Wait 0.3 s]     Wait03_1 --&gt; Pin5Off[/pin 5 off/]     Pin5Off --&gt; Wait03_2[Wait 0.3 s]     Wait03_2 --&gt; Pin1On{pin 1 on?}     Pin1On -- No --&gt; Pin5On     Pin1On -- Yes --&gt; Pin0On_2{pin 0 on?}     Pin0On_2 -- No --&gt; Pin5On     Pin0On_2 -- Yes --&gt; Pin7Off[/pin 7 off/]     Pin7Off --&gt; Pin0On   </pre> | 9        | <p>Pin numbers must be correct where applicable.</p> <p>Pin 7 on and off positions - 1 mark.</p> <p>Pin 6 on and off positions - 1 mark.</p> <p>All three delays positions (0.8 s and 2 x 0.3 s) with unit - 1 mark.</p> <p>Pin 5 on and off in positions - 1 mark.</p> <p>Pin 1 on ? with Y/N - 1 mark.</p> <p>Loop and arrow (Pin 1 on ?) back to before Pin 5 on - 1 mark.</p> <p>Pin 0 on ? with Y/N loop and arrow - 1 mark.</p> <p>Continuous loop to start with arrow - 1 mark.</p> <p>All marked symbols correct - 1 mark.</p> <p>Ignore any additional steps.</p> |
|          | (b) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2        | <p>1 mark pin 6 connection.</p> <p>1 mark for buzzer symbol and connection.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

[END OF MARKING INSTRUCTIONS]